1. Select the theme

1.1 Validate Selection
1.2 Define The problem
1.3 Form a team
1.4 Make A project Plan

QC Story Project

Project Title:- To Eliminate the concern of discharge hose cut in Scorpio (140HP) model

Automotive Division, Nasik Plant , Scorpio / Xylo TCF Dept.

	Plan	Actual
Start Date	02.04.2019	02.04.2019
Completion Date	30.9.2019	30.12.2019

Current Status:

Team	Name	Dept
Leader	**Vikrant R Ghule.** Email:GHULE.VIKRANT@gmail.com	MQA-Scorpio / Xylo TCF
Members	Sumit Mahale	PVT
	Nirmal Smabre	SQE
	Rayappan Anandhan	HVAC system COE
Dept Head	**Rahul Narkar**	Scorpio / Xylo TCF
Mentor	Shrikant Mulay	

1.1 Validate Selection Risk Assessment

No.	Success predictors	Yes (2)	Partial (1)	No (0)
1	**Project is a strategic priority, aligned organization goal (Big Y)**	2		
2	**Identify project area of focus :** Be sure there is no overlap with other MYB/MGB/MBB projects	2		
3	**Scope of the project is under the area of my influence**	2		
4	**Project scope is limited to only one product / one type of defect**	2		
5	**It is related to repetitive problem (chronic, not sporadic)**	2		
6	**Root cause is unknown and solution is unknown** (Type 4 problem)	2		
6	**Current level of the project is measured**	2		
7	**Potential Benefits of the project is identified**	2		
8	**Right vertical & horizontal team members are available**	2		
9	**Team members include technical experts**	2		
10	**Management is willing to commit serious resources to solutions**	2		
11	**Project is capable of completion within 4~6 months**	2		
12	**Project registration form is approved by HOD**	2		
	*** Any factor less than "2" creates a risk**			
			Total:24	

Success probability : High=22~24 Medium=19~21 Low=1~18

1.2 Define the problem

2. 4W 1H of the problem:

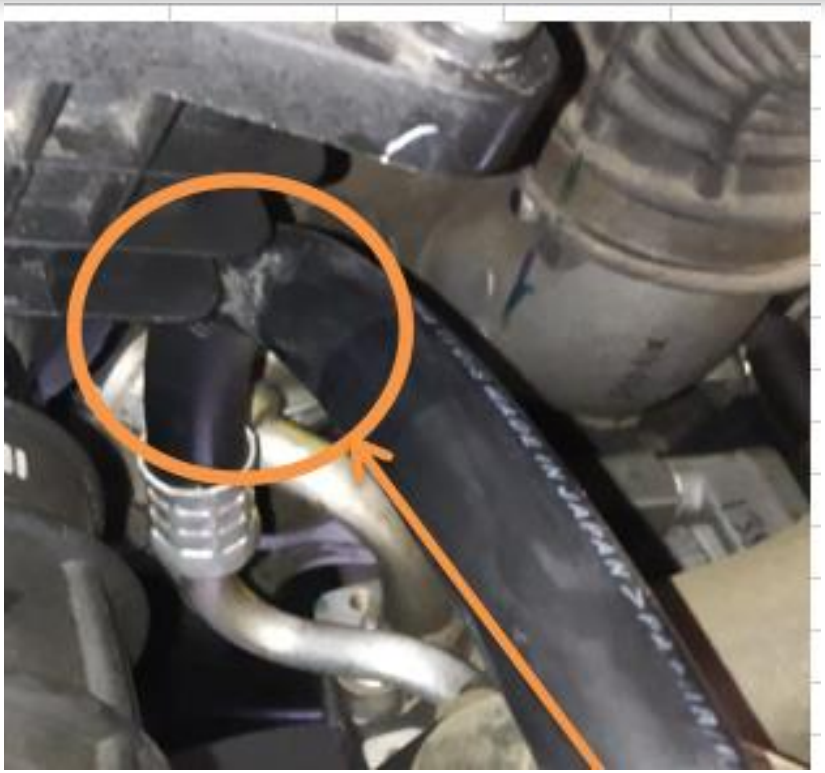
5W 1H	Details
What	**To Eliminate the concern of discharge hose cut in Scorpio (140HP) model**
When	While engine movement Scorpio (140HP) model. **Since start of vehicle production**.
Where	In field.
Who	Reported by the Customer.
How Big	Last 12 month 42 cases reported for discharge hose cut.

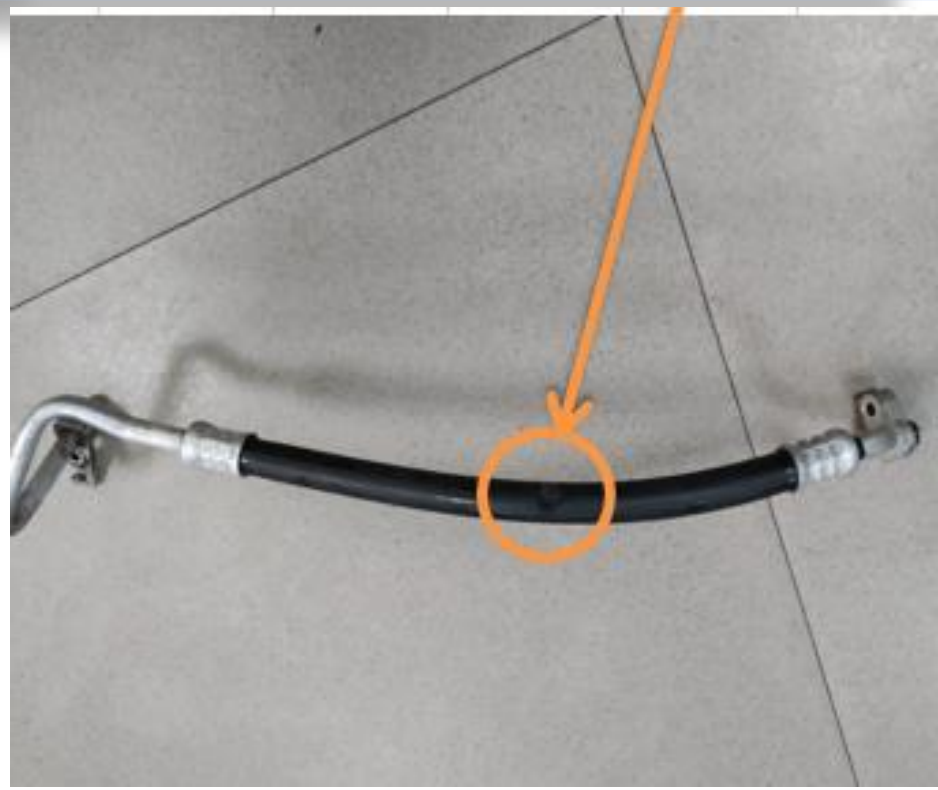
1.2 Define the problem

1.Brief Description of the problem:

To Eliminate the concern of discharge hose cut in Scorpio (140HP) model

Defect :-
Discharge hose cut due to fouling with air cleaner.





1.2 Define the problem

3. Problem details

S.NO	Check points	Details
1	Vehicle Model Selected for project	Scorpio – 140HP Models
2	Assembly / Part selected for project	Part Selected for Project :- 1) Discharge hose 2) Air cleaner 3) Engine 4) Condenser
3	Other Similar Models having the problem	No
4	Other similar models not having the problem:	Marazzo ,XYLO.
5	Stages where the problem can be detected	Stage, FAI , CAI buy off.
6	Processes under consideration (boundary/scope of project)	Supplier Parts- Discharge hose. Discharge hose fitment
7	Response type (Y):Variable/Attribute.	Attribute
8	Specification of the response:	Discharge hose fouling
9	Least count of Measurement System (if applicable)	NA
10	Whether MSA is needed? Yes/No	No

1.3 Form Team (involvement Matrix)

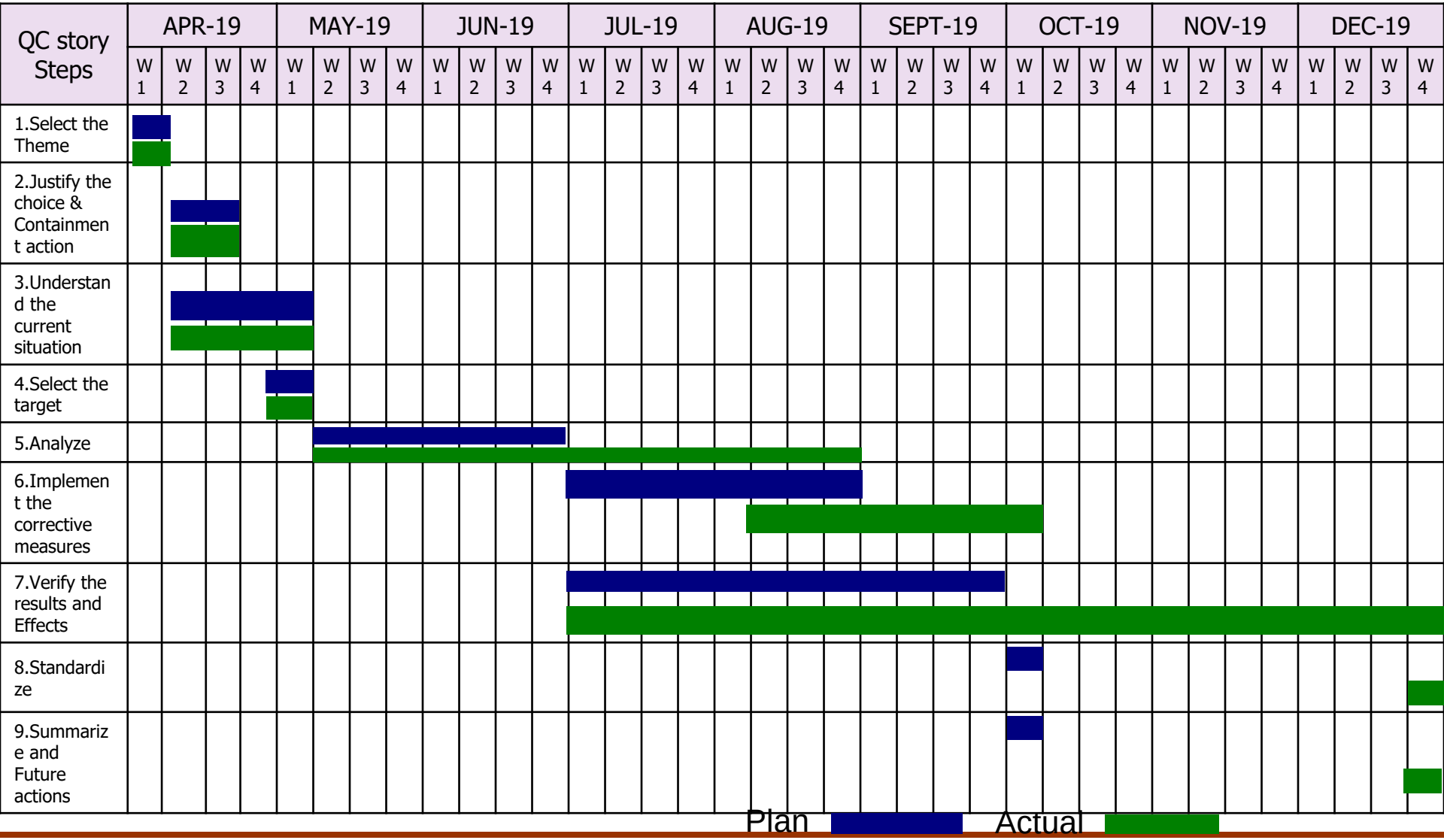
Instructions: Use the matrix to help you identify the people who should be involved in identifying cause, developing and implementing solutions.

Task	Details			
	Responsible for	Involved in	Consulted with	Informed About
Identifying causes	CFT	PVT & SQE	SQE, Supplier & Body shop	Supplier
Validating the causes	MQA			
Identifying Solutions	MQA, PVT & SQE			
Selecting Solutions	MQA , SQE & Supplier	MQA, PVT& SQE		Supplier
Planning Implementation	SQE & Supplier	MQA, & CDMM	Supplier	
Handling potential problems	SQE	PVT, MQA	Supplier	
Implementing solutions	Supplier & Bodyshop	PVT, SQE		
Monitoring results	MQA	PVT		PVT

Do we need involve anymore person in team? NO

1.4 Project Plan

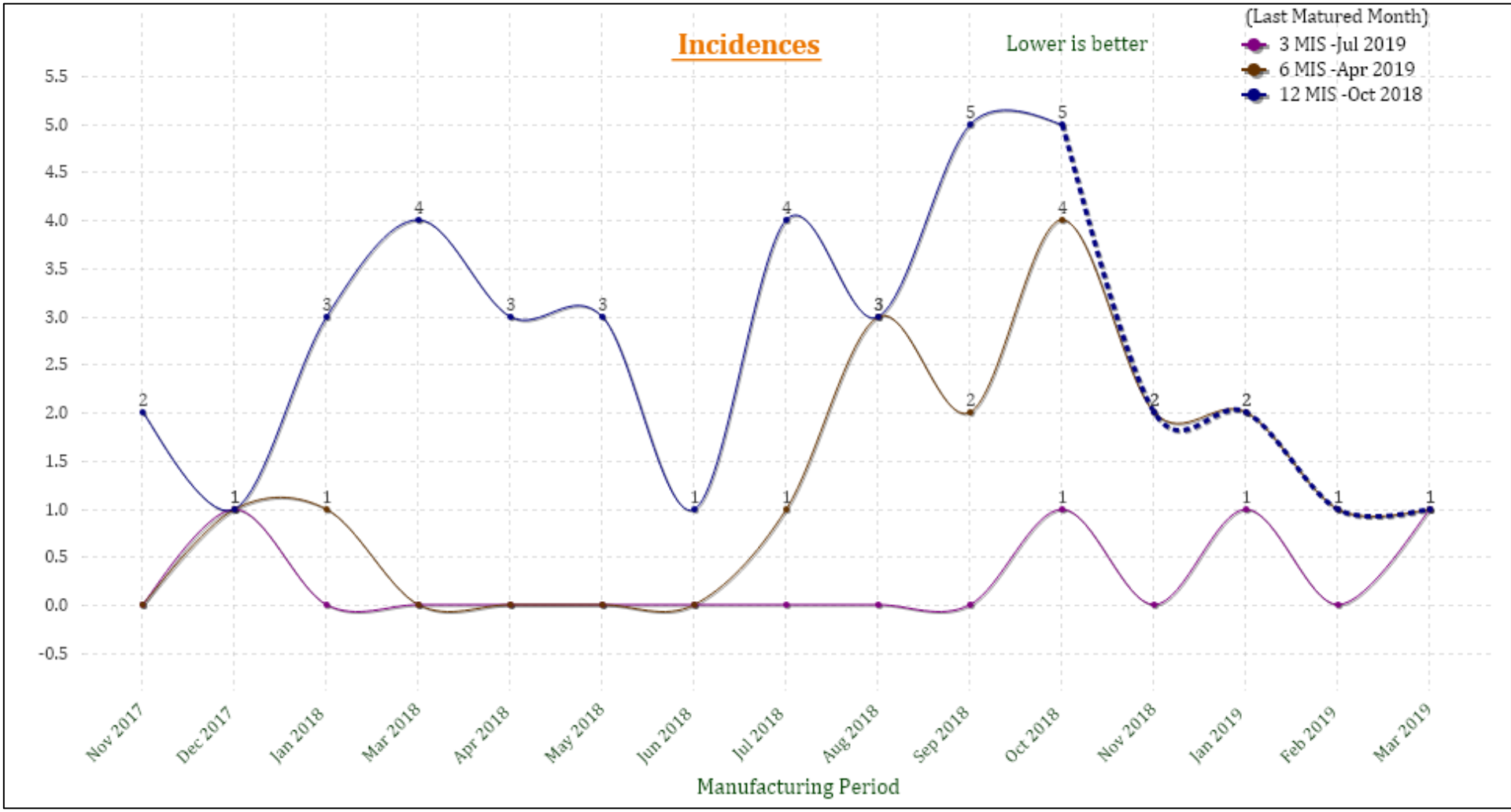
Project Planning



2 Justify the choice and containment action

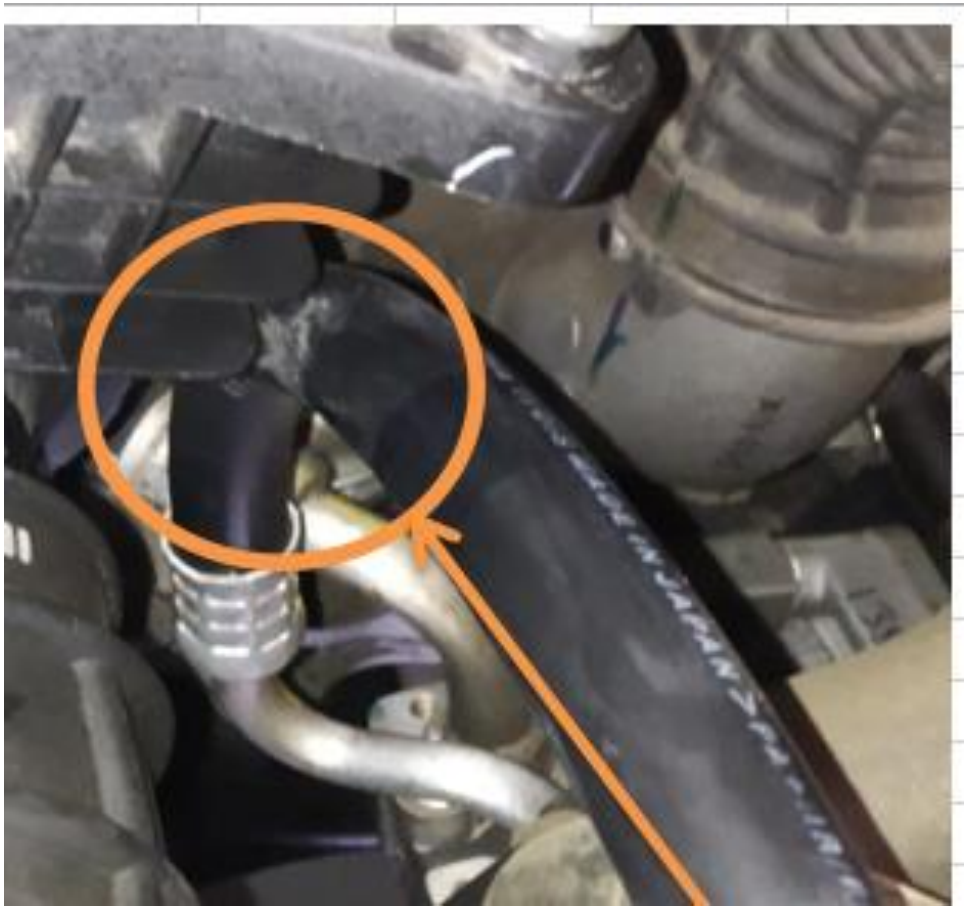
2.1 Past trend of the problem
2.2 Pareto Diagram
2.3 Impact of the problem
2.4 Cost of Poor Quality
2.5 Containment action

2.1 Past Trend of the problem



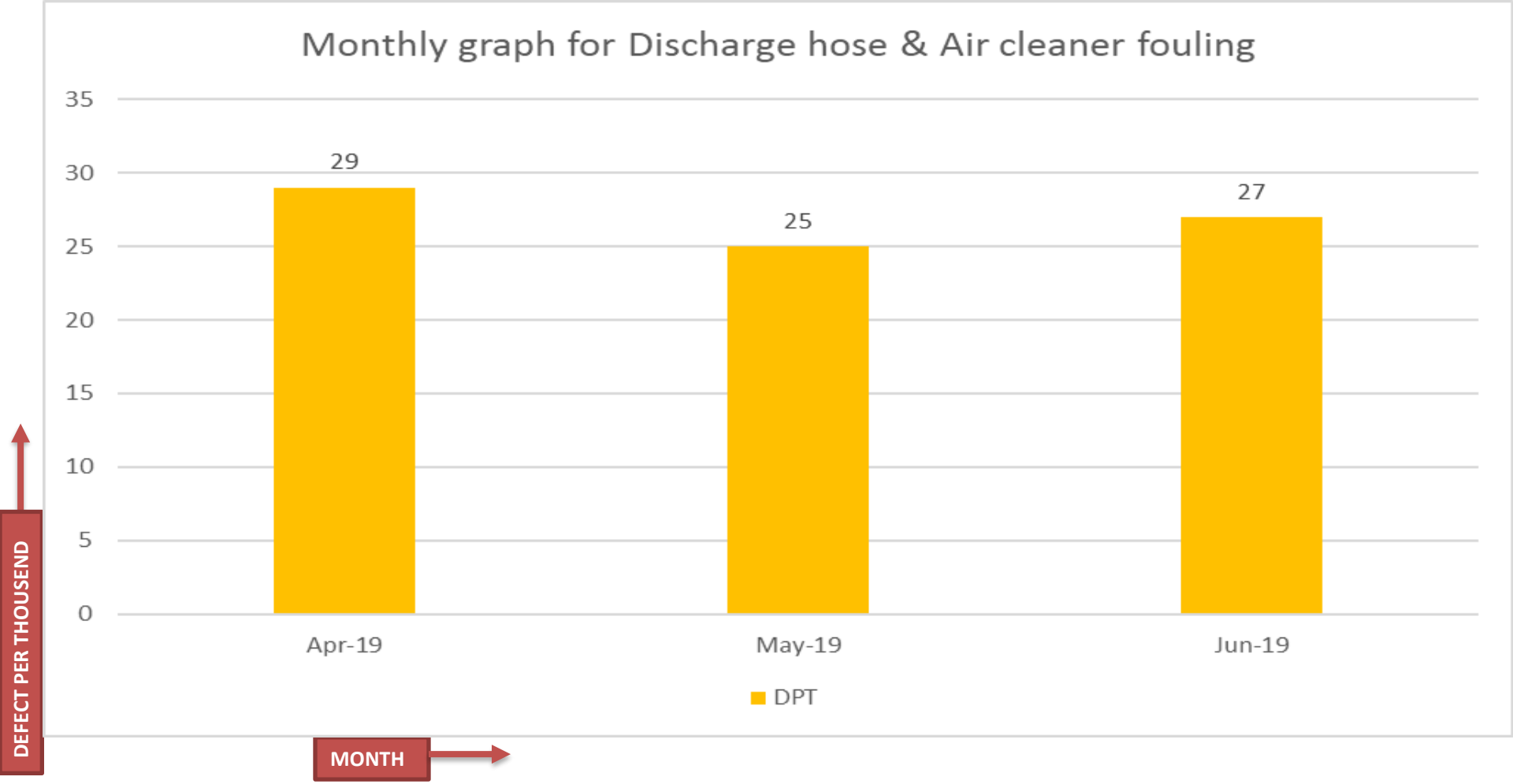
Inference:- AC hose cut/leakage is affecting 3, 6 & 12 MIS.

2.1 Past Trend of the problem



For customer protection
We added check point addition at roll down for touching.

2.1 Past Trend of the problem



Inference:- AC hose fouling to air cleaner defect is generated inhouse. This gets elevated to hose cut in field in 3,6,12 MIS.

Linkage with sector level goal

1. Tree Diagram

Linkage with sector level goal

Sector Objective :-
- J D Power score
Departmental Objective :-
– 3MIS Improvement
- Customer satisfaction

2.3 Impact of the problem

Impact on customer

1. Vehicle primary function not working
2. Major issue will be faced when AC not working.
3. Customer Dissatisfaction from functional point of view.

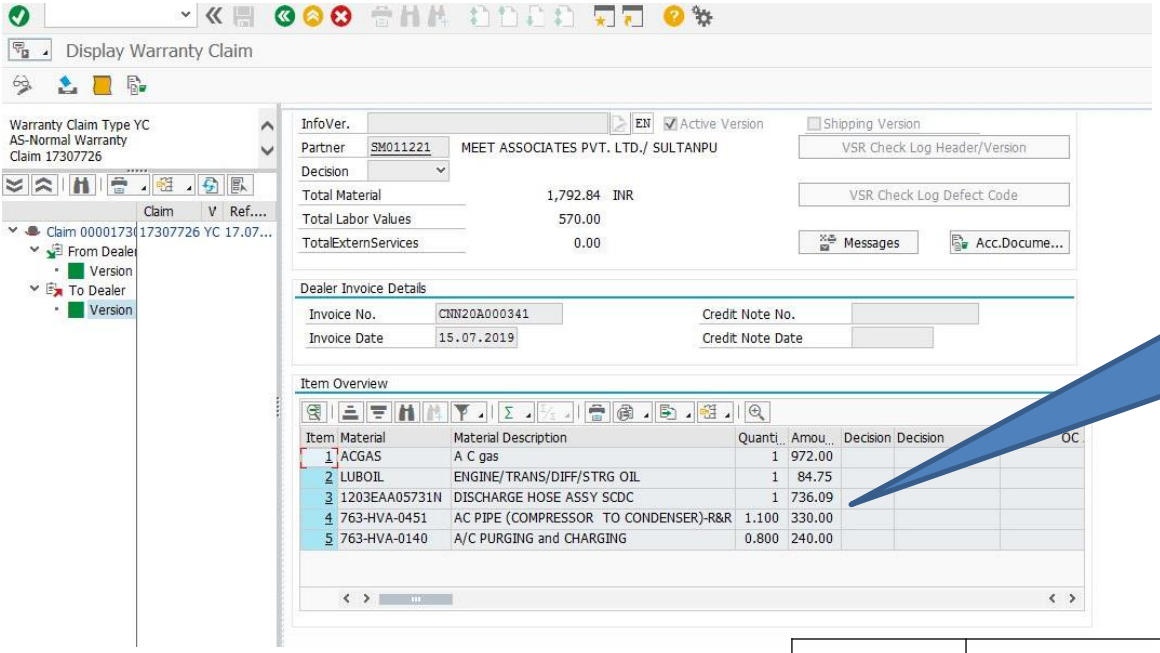
Impact on Mahindra brand

1. Customer Delightful experience with Scorpio Brand affected.
2. Promise 2020 affected.

Impact on Department goal

1. 3,6,12 MIS score will be effected.
2. Quality indicators effected
3. Claim cost increased.

2.4 Cost of Poor Quality



SAP Claim cost of one failure from dealer

Sr no	Material	Cost (Rs)
1	A/C Gas	972
2	Compressor Oil	84.75
3	Discharge hose assy SCDC	736.09
4	A/C pipe (Compressor to Condenser) R&R	330
5	A/C purging and charging	240
	Total	**2362.84**

Total 42 incidences in one year	**99,239**

2.4 Cost of Poor Quality

Summary of total benefits expected from the project:

1. Customer complaint resolution.
2. Improvement in Quality indicators
   (Customer complaint elimination)
2. Cost saving of **Rs.** 99239/- per year – as per COPQ
3. Quality Crusade targets improvement by First time right , Every time right & Customer delight

2.5 Containment Action

- Awareness DRF, Yard check for defect.

1.DRF given to all stage associates to make aware about defect and protect customer

2.Yard check done to ensure already build vehicle with defect free condition

3.Check point addition at roll down for touching.

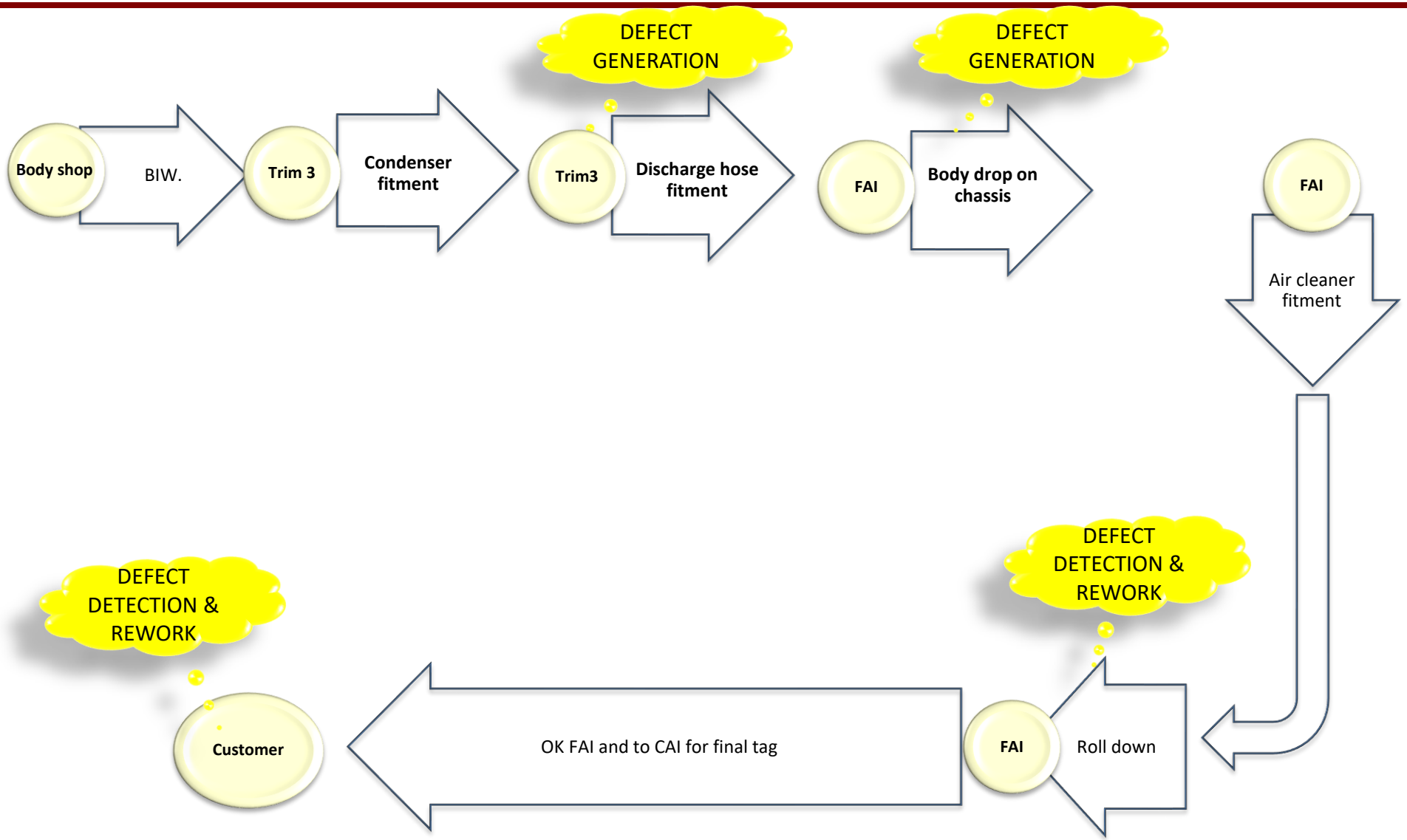
<u>Rework of defective vehicle</u>

- Defective vehicle reworked by changing pipe orientation on condenser side

3. Understand the current Situation

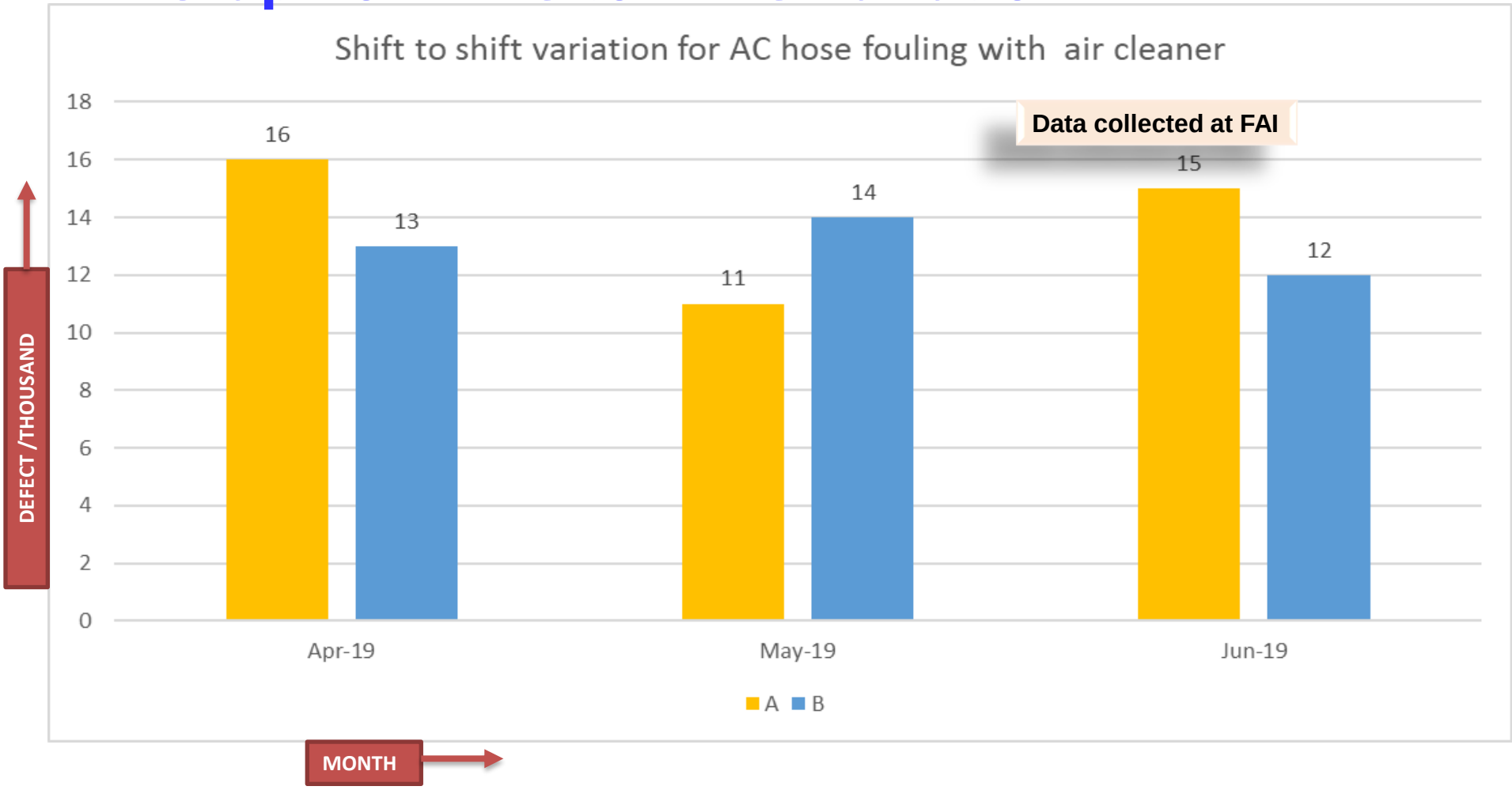
3.1 Process Flow Diagram
3.2 Past data Analysis from different view point
3.3 Gemba Observations
3.4 Quick win Identification
3.5 Results of Quick wins

3.1 Process Mapping



3.2 Different View Point

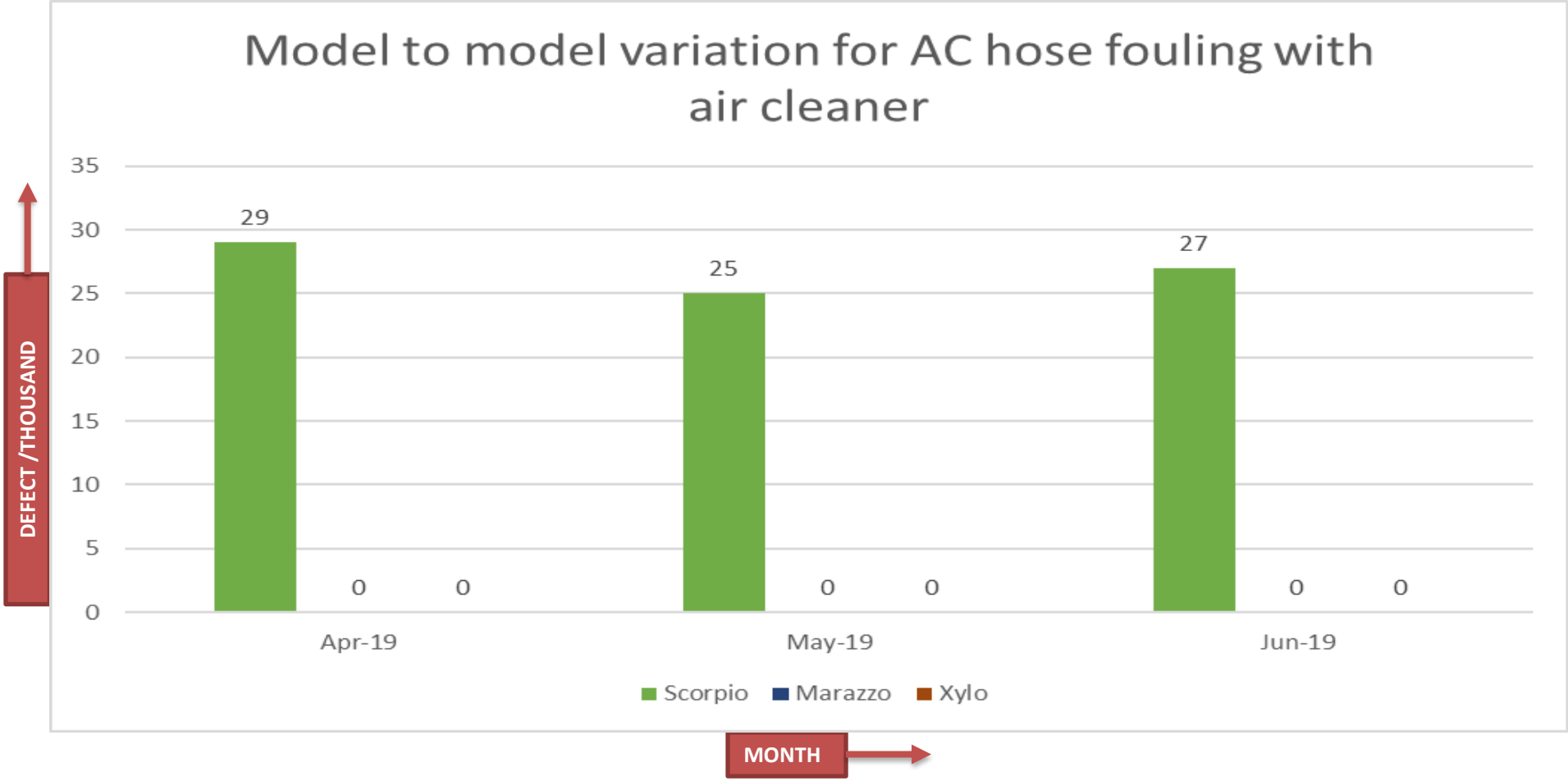
A. Graph of Time to Time Variation:



Inference :- No significant relation established.

3.2 Different View Point

B. Stream to Stream



Inference :- Scorpio model is affected badly because of defect.

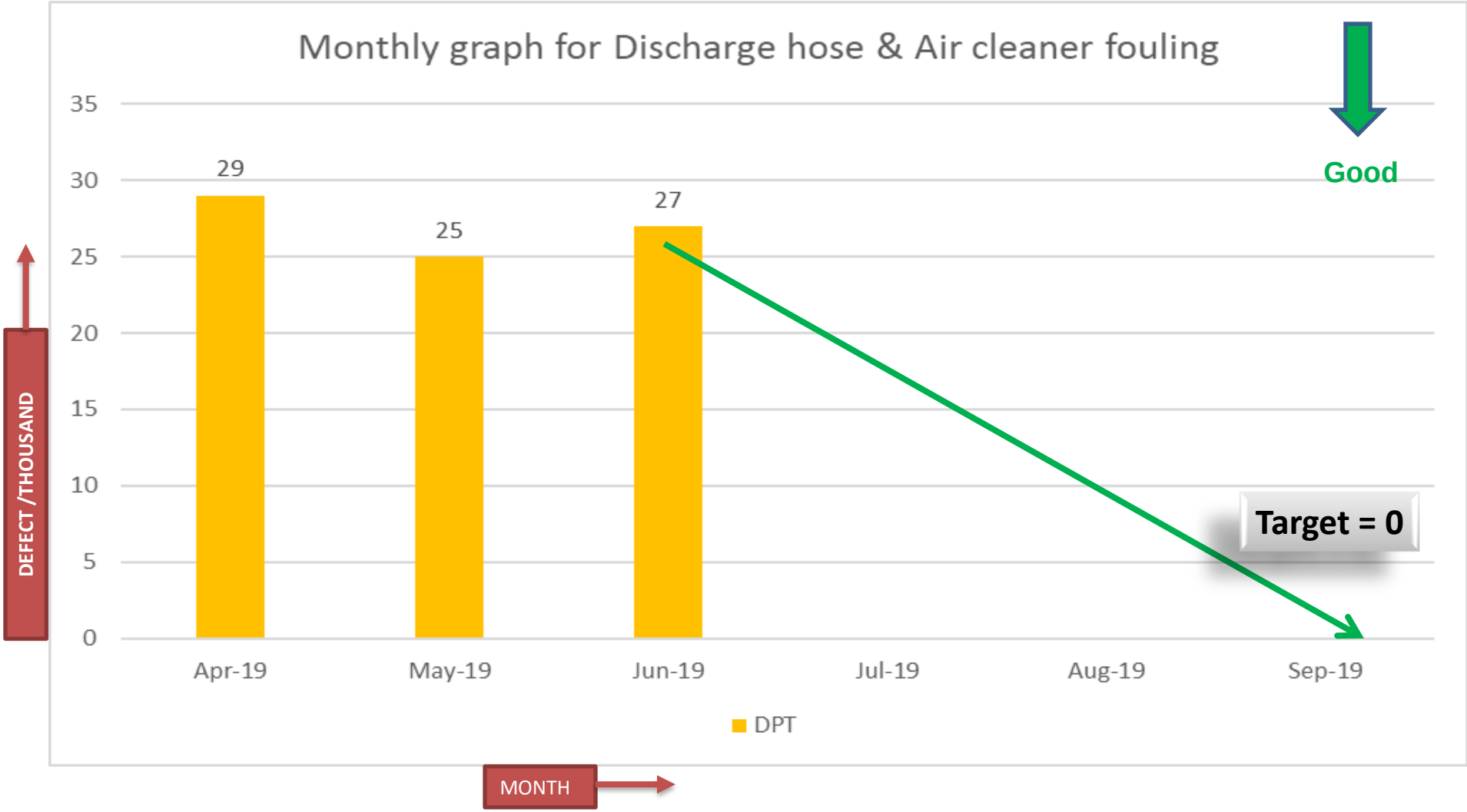
3.3 Gemba Observations

Observations from Past data analysis/Failed part/Gemba visit

1. 4M (Part 1)- LO Analysis done – Process Conforming
2. Battery tool condition found ok (6±1 Nm Torque)
3. Part condition found ok (No pipe bend in storage)
4. Trained associate (Black L)
5. Associate follows process as per SOS

No quick wins identified during Gemba observations

4. Select the target



Inference : Target of defect elimination by Dec is taken.

5. Analyze

5.1 List the SSVs

5.2 Identify the tool

5.3 Collect the data

5.4 Test the Hypothesis

5.5 Conclusions of Analysis

- *Decide funneling needed or not*

5.6 Funneling to Root Cause

5.1 List down SSV's or Xs

$$Y = f(x)$$

Response SSV

Y = Discharge hose fouling to air cleaner.

X = X1. Assembly process X2. Components in the assembly

- Initially CFT decided to find the root cause from Assembly process (X1)

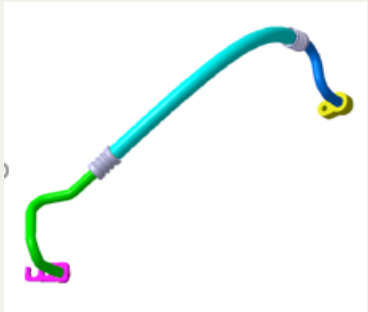
SIX SIGMA Tool identified

"COMPONENT SEARCH" or "MODIFIED COMPONENT SEARCH"

Based on assembly function

5.1 List down SSV's or Xs

- Team listed the parts in descending order, in which it will affect the response

Part Description	1	2	3	4
	Air cleaner	Discharge hose	Condenser	Engine
IMAGES				

5.2 Identify the tool

- Team listed the parts in descending order, in which it will affect the response

Sr. No	Part Description	Usage of parts after disassembly
1	Air cleaner	Can be reused
2	Discharge	Can be reused
3	Condenser	Can be reused
4	Engine	Can be reused

Hence DOE Tool
“COMPONENT SEARCH"
was selected

5.2 CRITERIA FOR SCALING – Y1 & GRAPH

- Team listed the parts in descending order, in which it will affect the response

Response (Y) – **Discharge hose foul**
SSV (X) - Parts combinations & assembly process.

Response scale:-

1 = OK – Discharge hose not fouling

- 1 = NOT OK – Discharge hose fouling to air cleaner

5.2 Analysis

Shainin DOE –Component Search – Graphical Presentation :-

A) Assembly Process Confirmation

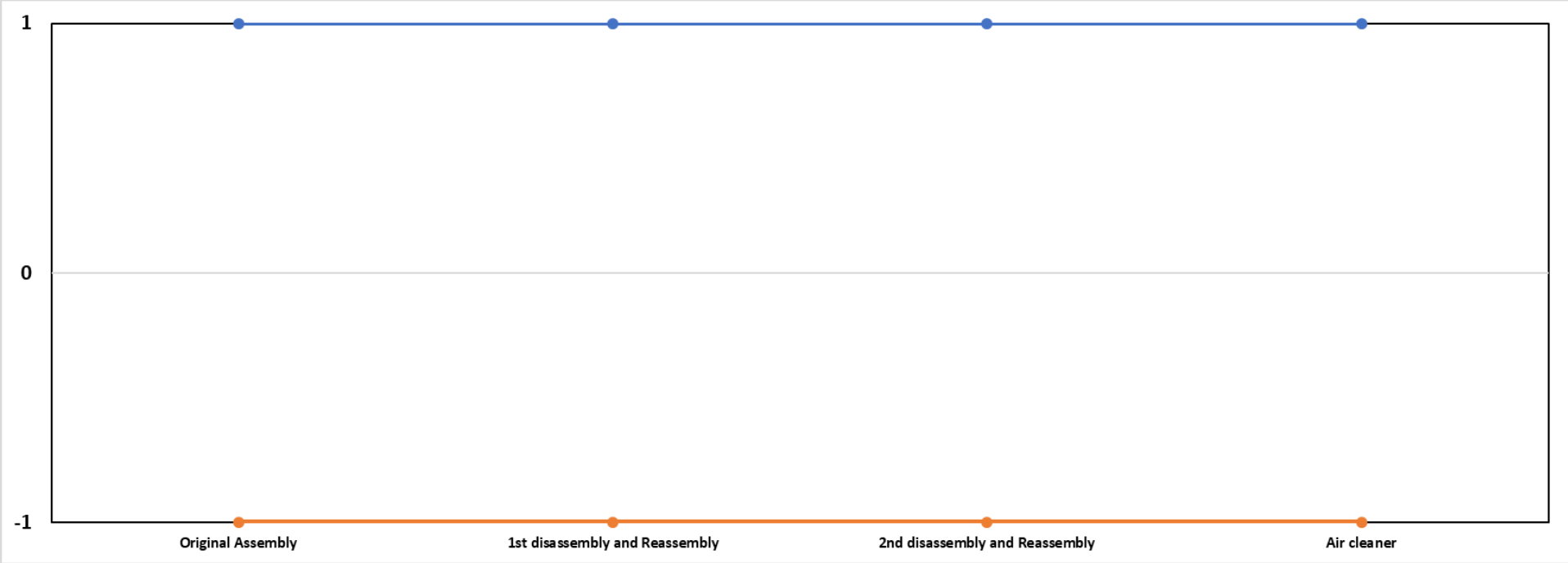
Inference :- Assembly Process is not a cause.

While assembly and dis assembly of sub components OK vehicle remains ok and not ok vehicle remains Not ok.

5.2 Analysis

Shainin DOE –Component Search – Graphical Presentation :-

B) Component contribution confirmation:-



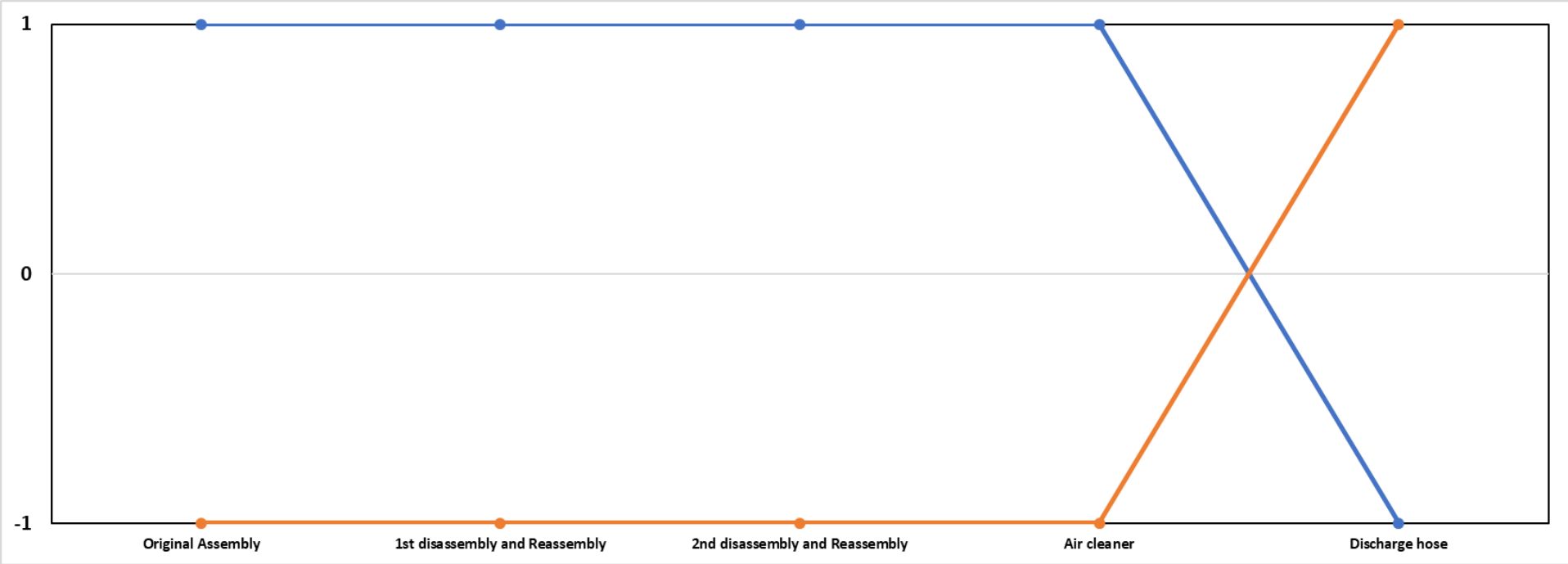
Inference :- Air cleaner is not a cause.

While air cleaner swapping- OK vehicle remains ok and not ok vehicle remains Not ok.

5.2 Analysis

Shainin DOE –Component Search – Graphical Presentation :-

B) Component contribution confirmation:-



Inference :- Discharge hose is a cause.

While discharge hose swapping- OK vehicle became not ok and not ok vehicle remains OK.

5.1 List down SSV's or Xs

$$Y = f(x)$$

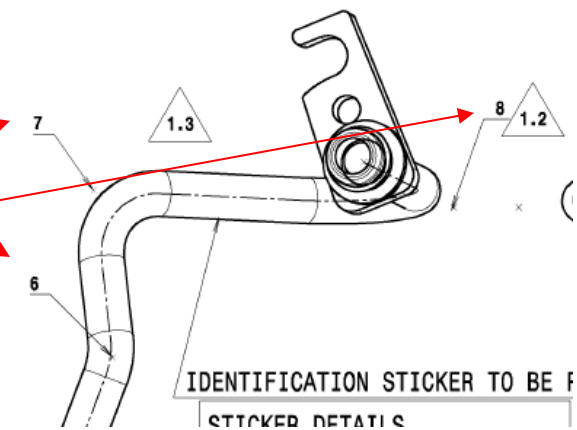
Response **SSV**

Y = Discharge hose fouling

(i.e BOB and WOW respectively)

X = Discharge hose causing the defect.

X1- Hose length after crimping
X2- Bend angle – Compressor end
X3- Bend angle – Condenser end -1
X4- Bend angle – Condenser end -2
X5- Bend angle – Condenser end -3
X6- Connector angle



5.2 Identify the tool

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X1	Hose length after crimping	350 +10 mm	**Paired Comparison**
X2	Bend angle – Compressor end	93.31 +/- 2 Deg	**Paired Comparison**
X3	Bend angle – Condenser end -1	31.22 +/- 2 Deg	**Paired Comparison**
X4	Bend angle – Condenser end -2	89.88 +/- 2 Deg	**Paired Comparison**
X5	Bend angle – Condenser end -3	130.47 +/- 2 Deg	**Paired Comparison**
X6	Connector angle	91.9 +/- 2 Deg	**Paired Comparison**

The identified SSV's are Measurable on both Good and Bad components and are Input related Materials Also parameters can not be loss during measurement or not damage.

Hence DOE Tool

"Paired Comparison"

was selected.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X1	Hose length after crimping	350 +10 mm	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X1
W1	360
W2	356
W3	357
W4	359
W5	355
W6	350
W7	356
W8	355
B1	350
B2	357
B3	358
B4	355
B5	355
B6	356
B7	354
B8	356

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X1	Hose length after crimping	350 +10 mm	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X1
W6	350
B1	350
B7	354
W5	355
W8	355
B4	355
B5	355
W2	356
W7	356
B6	356
B8	356
W3	357
B2	357
B3	358
W4	359
W1	360

Inference:-

Due to rule 1 this is not cause

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X2	Bend angle – Compressor end	93.31 +/- 2 Deg	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X2
B1	95.3
B2	93.3
B3	93.1
B4	93.2
B5	94
B6	94.9
B7	94.7
B8	95.3
W1	92.1
W2	93.1
W3	94.1
W4	92.1
W5	94.1
W6	95
W7	93.1
W8	94.2

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X2	Bend angle – Compressor end	93.31 +/- 2 Deg	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X2
W4	92.1
W1	92.1
W2	93.1
W7	93.1
B3	93.1
B4	93.2
B2	93.3
B5	94
W5	94.1
W3	94.1
W8	94.2
B7	94.7
B6	94.9
W6	95
B1	95.3
B8	95.3

Inference:-

Due to rule 1 this is not cause

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X3	Bend angle – Condenser end -1	31.22 +/- 2 Deg	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X3
W1	26.2
W4	28.8
W3	28.9
W2	27.8
W5	28.2
W6	29.1
W7	31.2
W8	29.4
B1	32.1
B2	33.2
B3	30.1
B4	31.2
B5	32.1
B6	33.1
B7	31.2
B8	32.4

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X3	Bend angle – Condenser end -1	31.22 +/- 2 Deg	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X3
W1	26.2
W2	27.8
W5	28.2
W4	28.8
W3	28.9
W6	29.1
W8	29.4
B3	30.1
W7	31.2
B4	31.2
B7	31.2
B1	32.1
B5	32.1
B8	32.4
B6	33.1
B2	33.2

TOP COUNT =7

BOTTOM COUNT =7

Inference:-

Top count – 7
Bottom count – 7
Total – 14

Total 14 which is more than 8. hence Bend angle – Condenser end -1 less is possible cause.

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X4	Bend angle – Condenser end -2	89.88 +/- 2 Deg	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X4
W1	92
W4	92.1
W3	92.01
W2	91.9
W5	92.2
W6	92.2
W7	92
W8	91
B1	89.8
B2	91.5
B3	91.4
B4	91.8
B5	91.8
B6	90.1
B7	90.9
B8	90.7

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X4	Bend angle – Condenser end -2	89.88 +/- 2 Deg	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X4
B1	89.8
B6	90.1
B8	90.7
B7	90.9
W8	91
B3	91.4
B2	91.5
B4	91.8
B5	91.8
W2	91.9
W1	92
W7	92
W3	92.01
W4	92.1
W5	92.2
W6	92.2

TOP COUNT =4

BOTTOM COUNT =7

Inference:-

Top count – 4
Bottom count – 7
Total – 11

Total 11 which is more than 8. hence Bend angle – Condenser end -2 more is possible cause.

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X5	Bend angle – Condenser end -3	130.47 +/- 2 Deg	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X5
B1	129.8
B2	129.9
B3	131.5
B4	131.9
B5	132.3
B6	132.2
B7	132.3
B8	132.4
W1	134.1
W2	133.2
W3	132.5
W4	135.6
W5	132.8
W6	132.9
W7	132.7
W8	133.1

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X5	Bend angle – Condenser end -3	130.47 +/- 2 Deg	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X5
B1	129.8
B2	129.9
B3	131.5
B4	131.9
B6	132.2
B5	132.3
B7	132.3
B8	132.4
W3	132.5
W7	132.7
W5	132.8
W6	132.9
W8	133.1
W2	133.2
W1	134.1
W4	135.6

TOP COUNT =8

BOTTOM COUNT =8

Inference:-

Top count – 8
Bottom count – 8
Total – 16

Total 16 which is more than 8. hence Bend angle – Condenser end -3 more is possible cause.

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.3 Collect the data

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X6	Connector angle	91.9 +/- 2 Deg	**Paired Comparison**

We have identified **8 BOB** - OK vehicle for not fouling hose and 8 **WOW** - NOT OK vehicle in which hose is fouling

Veh.	X6
B1	90.8
B2	91.3
B3	91.5
B4	91
B5	92.3
B6	92.5
B7	92.8
B8	93.9
W1	89.9
W2	92
W3	90.1
W4	90.5
W5	90.6
W6	90.7
W7	90.8
W8	90.7

5.4 Test the Hypothesis

Sr.No	SSV (X)	Drawing specifications	DOE / tool selected
X6	Connector angle	91.9 +/- 2 Deg	**Paired Comparison**

Data arrange in ascending order :-

Veh.	X6
W3	90.1
W4	90.5
W5	90.6
W6	90.7
W8	90.7
B1	90.8
W7	90.8
B4	91
B2	91.3
B3	91.5
W2	92
B5	92.3
B6	92.5
B7	92.8
W1	93.9
B8	93.9

Inference:-

Due to rule 1 this is not cause

Rule 1 – Highest & lowest values not in same category

Rule 2 – Highest & lowest values are repeat and in same category.

5.5 Conclusions of Analysis

X	Probable Cause	Valid/ Invalid	Any changes in Spec.	X Controllable as per new spec Y/N	Drilling down required Y/N
X3	Bend angle – Condenser end -1	Valid	No	YES	No
X4	Bend angle – Condenser end -2	Valid	No	YES	No
X5	Bend angle – Condenser end -3	Valid	No	YES	No

Conclusion :-

Best Condition – When condenser end1 angle is more, end 2 angle is less & end 3 angle is less

Worst Condition - When condenser end1 angle is less, end 2 angle is more & end 3 angle is more

5.6 Funneling to Next level causes

Discharge hose fouling to air cleaner.

Why ?

When condenser angle variation

Why ?

Supplier part variability outside drg. Specifications.

Why ?

Supplier not maintaining the parts within drgs spec.

Shining DOE confirms problem lies with 3 parameters.

ROOT CAUSE

HOW`

Concern to be handover to SQA dept. for bringing part within drawing specifications.

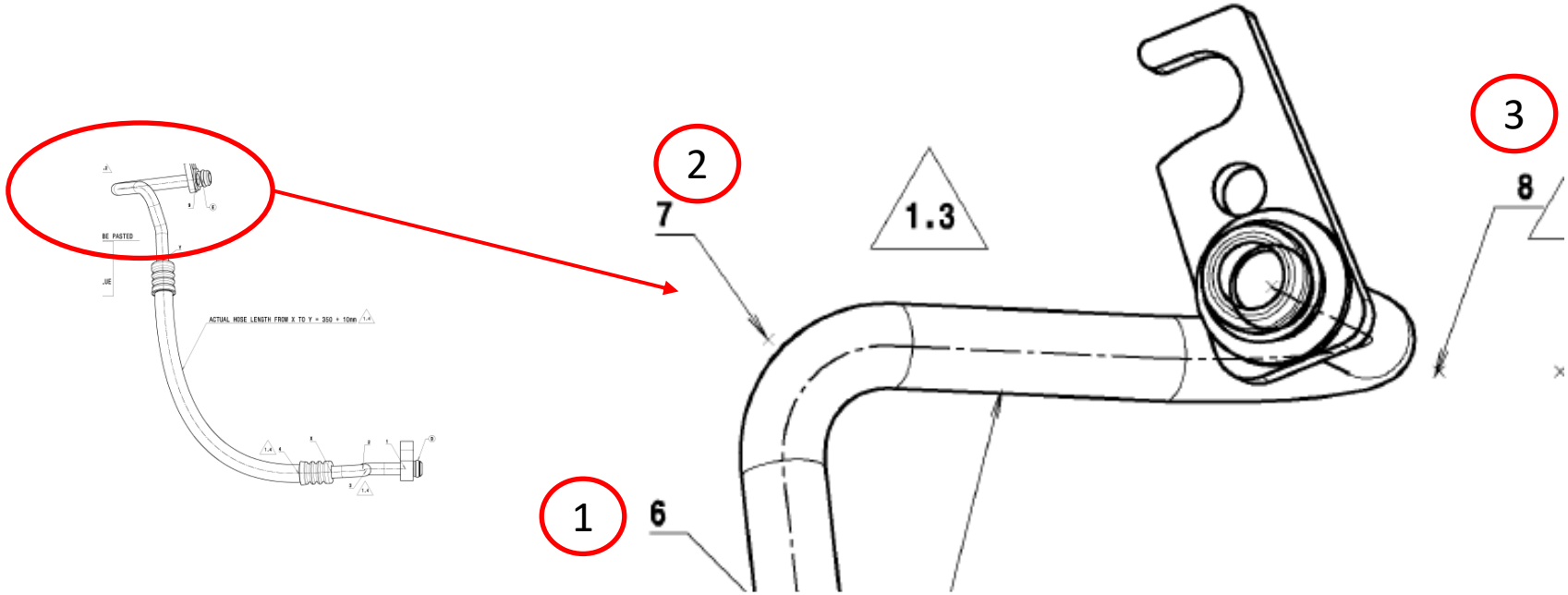
COUNTER MEASURE

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

6.1 Generate possible solutions

X	Validated causes	Possible Solutions
X3	Bend angle – Condenser end -1	Dimension must be maintained 31.22 +/- 2 Deg at supplier.
X4	Bend angle – Condenser end -2	Dimension must be maintained 89.88 +/- 2 Deg at supplier.
X5	Bend angle – Condenser end -3	Dimension must be maintained 130.47 +/- 2 Deg at supplier.



6.2 Evaluate and Select Solution

Solution Selection Matrix

		Effectiveness	Time Impact	Cost/Benefit Impact	<<<<Evaluation Criteria
		10	8	5	<<<<<<<<Importance
	Solution	Correlation of Solution to Criteria			Total
X3	Bend angle – Condenser end -1	10	8	7	560
X4	Bend angle – Condenser end -2	10	8	7	560
X5	Bend angle – Condenser end -3	10	8	7	560

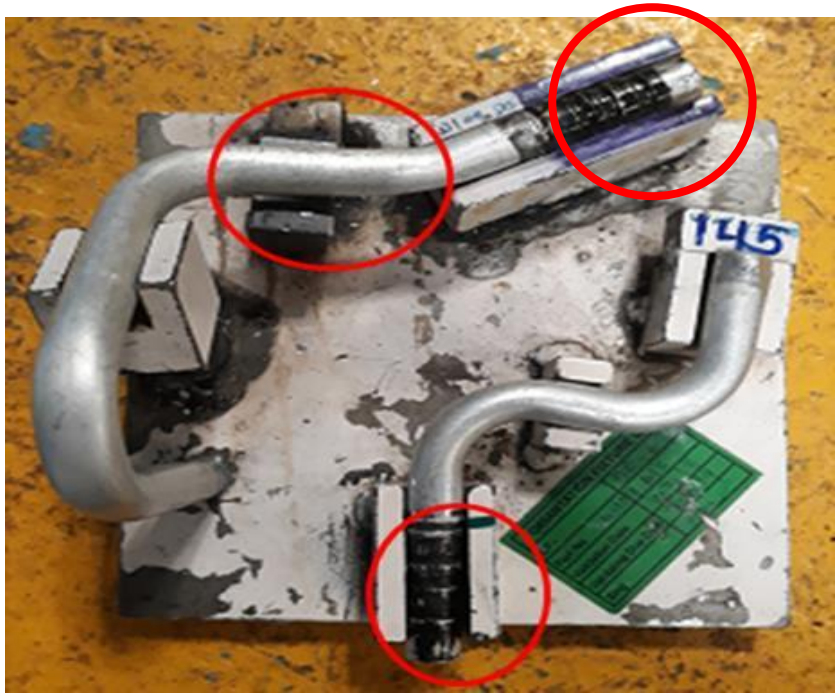
6.3 Implement Solution

Solution Implementation Plan

X	Validated causes	Actions finalized	Resp.	Target Date	Status
X3	Bend angle – Condenser end - 1	1. Bending programming done 2. Pipe Bending profile checking fixture corrected	Nirmal Sambre	12.Jul.2019	IMPLEMENTED
X4	Bend angle – Condenser end - 2				
X5	Bend angle – Condenser end - 3				

6.3 Implement Solution

Solution Implementation Plan



Pipe Not Qualifying and also control not available for Length

Before

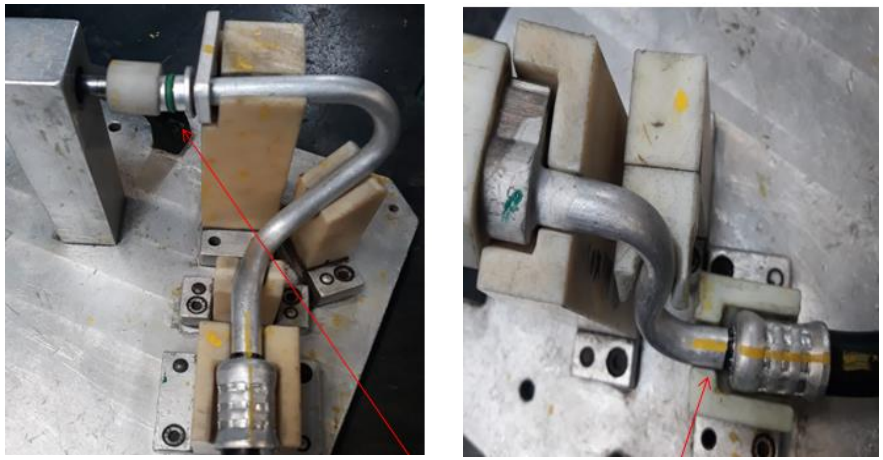
Pipe Profile Correction Done and Stopper Added for Length Control

After

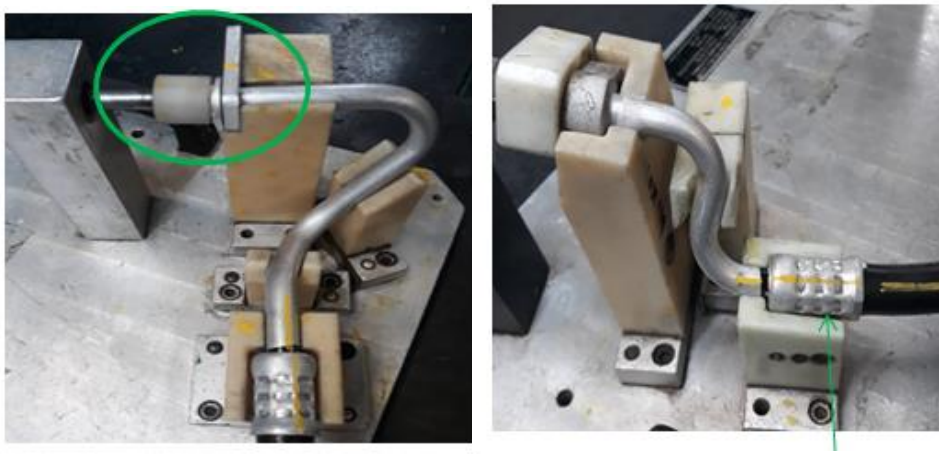
6.3 Implement Solution

Solution Implementation Plan

Final Profile Inspection:



Pipe Not Qualifying in Final Gauge



Pipe Qualification in Final Gauge after Bending programme correction

Before

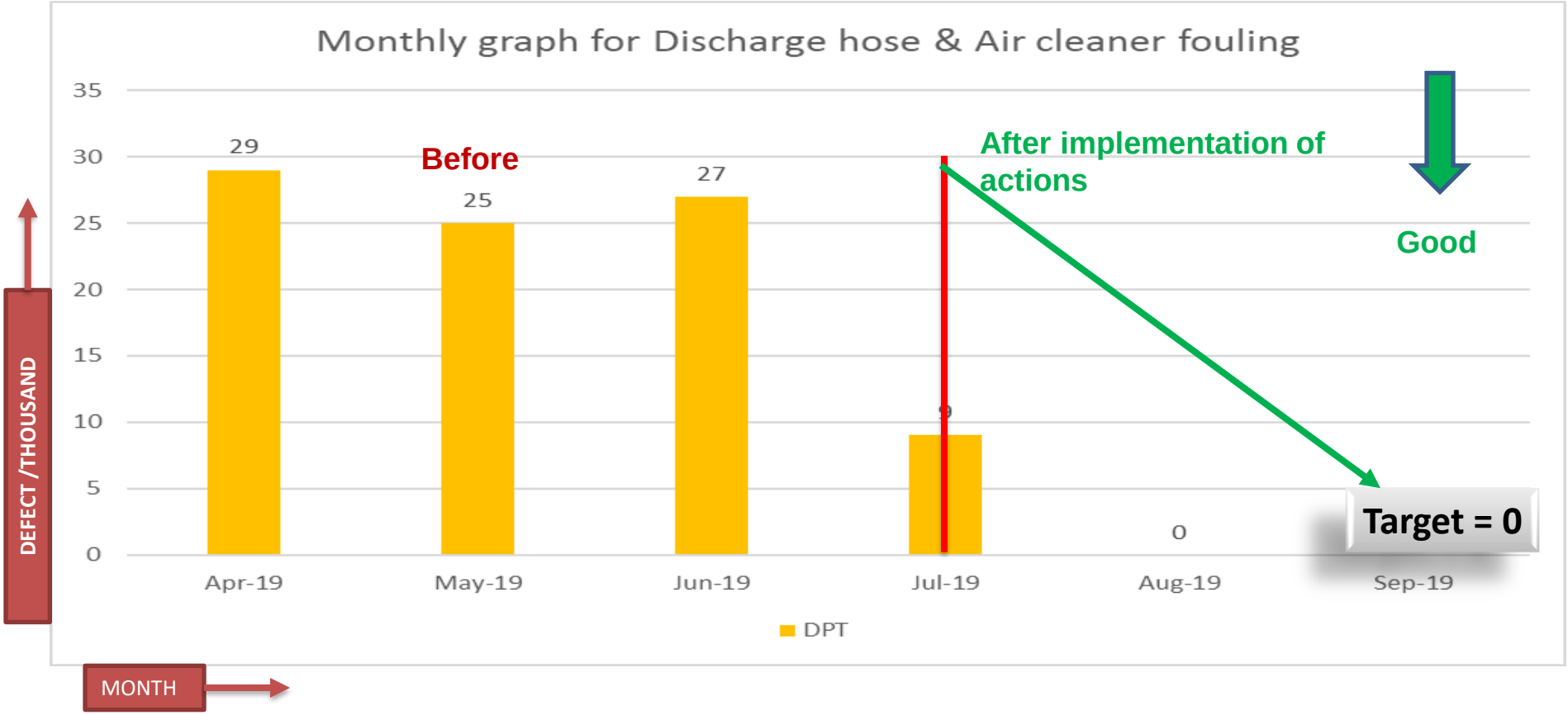
After

7. Confirm Results and effects

1. Collect data
2. Capture all effects
3. Compare before and after
4. Measure Benefits

7.1 Collect data

Data of after action – Month wise defect trend



Inference: **Discharge hose fouling to air cleaner defect has been eliminated after implementation of corrective actions**

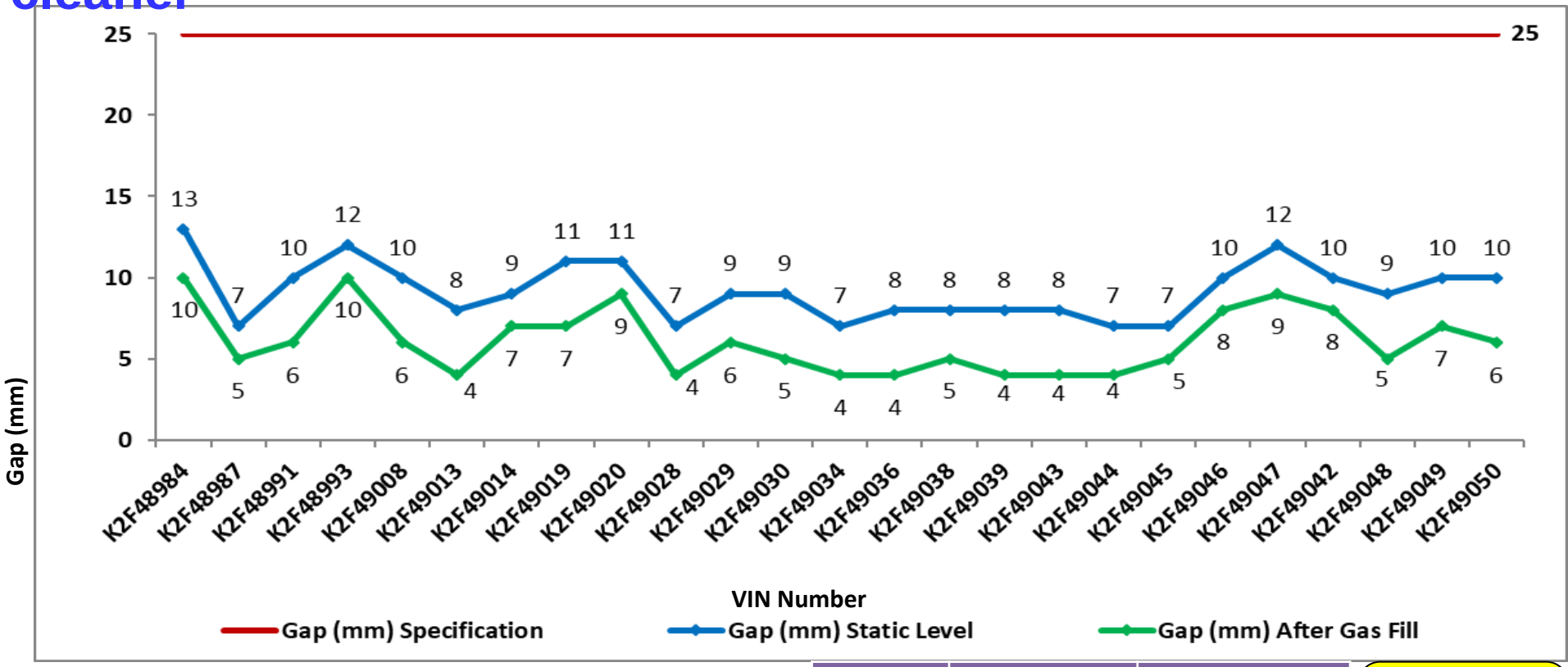
7.1 Collect data

Data of after action – Gap measured between hose and air cleaner

CAD clearances in existing layout is 25mm

7.1 Collect data

Data of after action – Gap measured between hose and air cleaner



Inference: Though fouling has been eliminated, still gap is very less as compared to specifications which may result in hose cut in field.

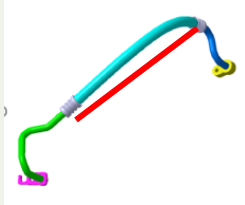
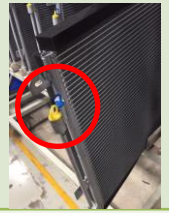
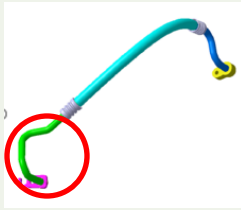
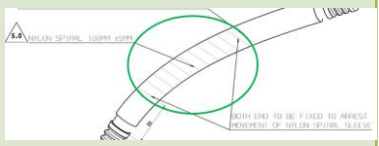
	Static Level	After Gas Fill
Average	9.2	6.1
Min	7	4
Max	13	10

After Gas Fill Gap is 4 to 10 mm

6. Implement Corrective Measures

1. Generate Possible Solutions
2. Evaluate and Select Solution
3. Implement the action

6.1 Generate possible solutions

Brain storming	Possible Solutions
Gap between AC pipe and air cleaner (<10mm)	Air cleaner ribs to cut
	AC hose length to be reduced 
	AC pipe mtg. on Condenser to be shifted 
	AC pipe bend angle condenser end to be modified 
	Spiral sleeve addition on hose 

6.2 Evaluate and Select Solution

Solution Selection Matrix

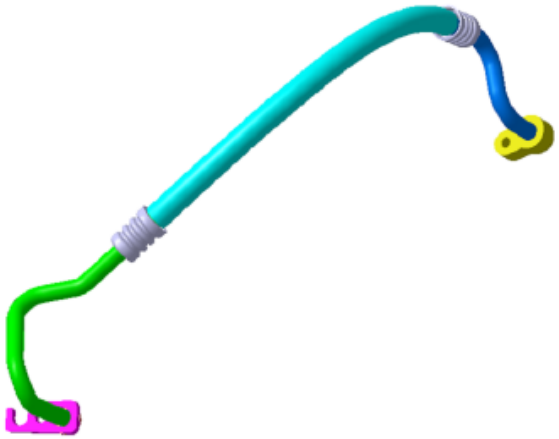
		Effectiveness	Time Impact	Cost/Benefit Impact	<<<<Evaluation Criteria
		10	8	5	<<<<<<<<Importance
	Solution	Correlation of Solution to Criteria			Total
1	Air cleaner ribs to cut	10	3	7	210
2	AC hose length to be reduced	10	4	5	200
3	AC pipe mtg. on Condenser to be shifted	10	2	1	20
4	AC pipe bend angle condenser end to be modified	10	8	8	640
5	Spiral sleeve addition on pipe	8	9	5	360

6.3 Implement Solution

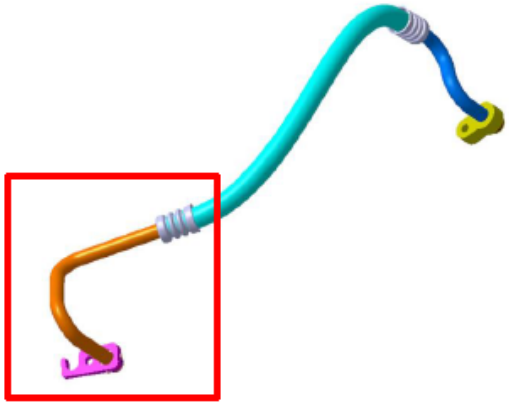
Solution Implementation Plan

X	Validated causes	Actions finalized	Resp.	Target Date	Status
X	Bend angle – Condenser end	Condenser bend angle modified to increase gap	Rayappan Anandhan	12.Jul.2019	IMPLEMENTED

Existing pipe_GX195001



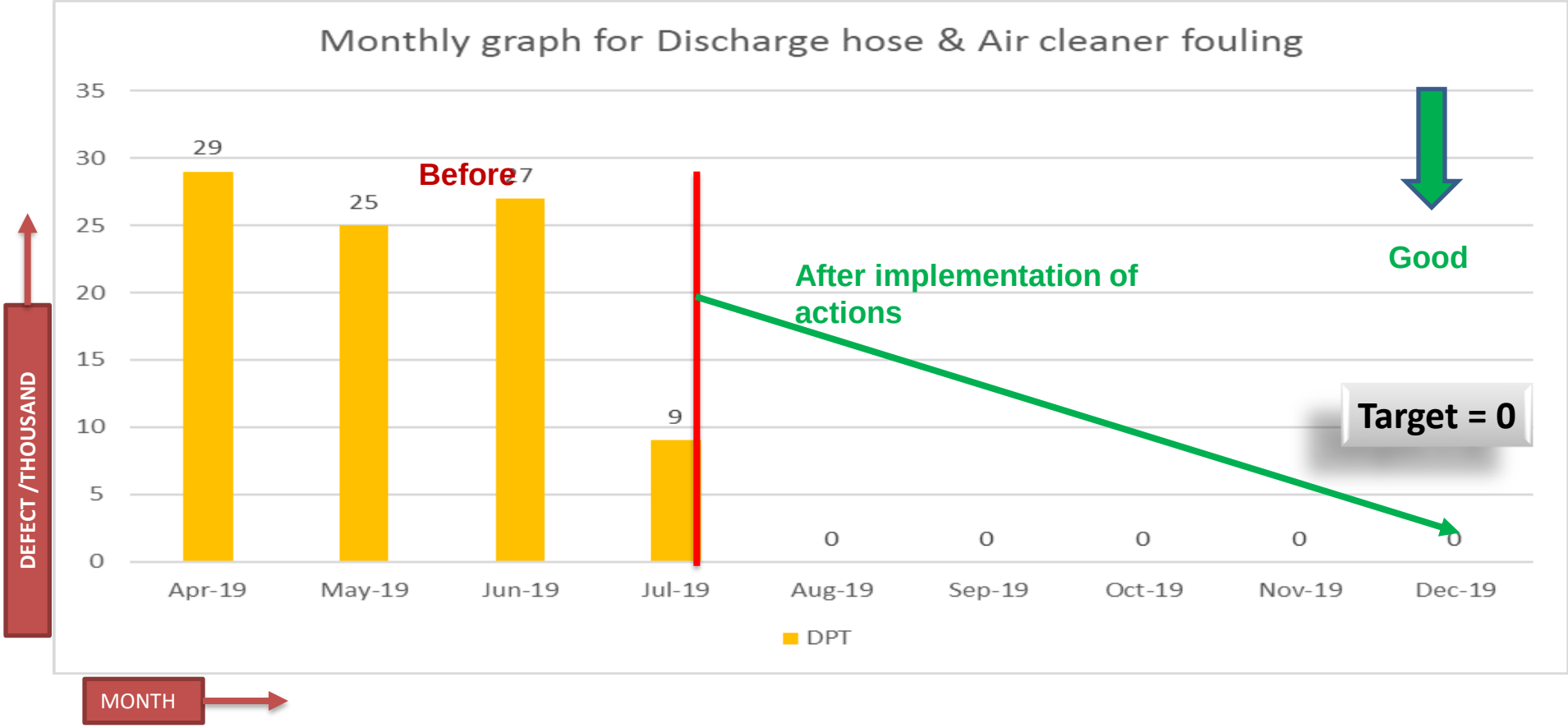
Modified pipe



Condenser end pipe routing modified as per vehicle trial.
Pipe modified in marked area,
Hose length remains same.

7.1 Collect data

Data of after action – Month wise defect trend



Inference: **Discharge hose fouling to air cleaner defect has been eliminated after implementation of corrective actions**

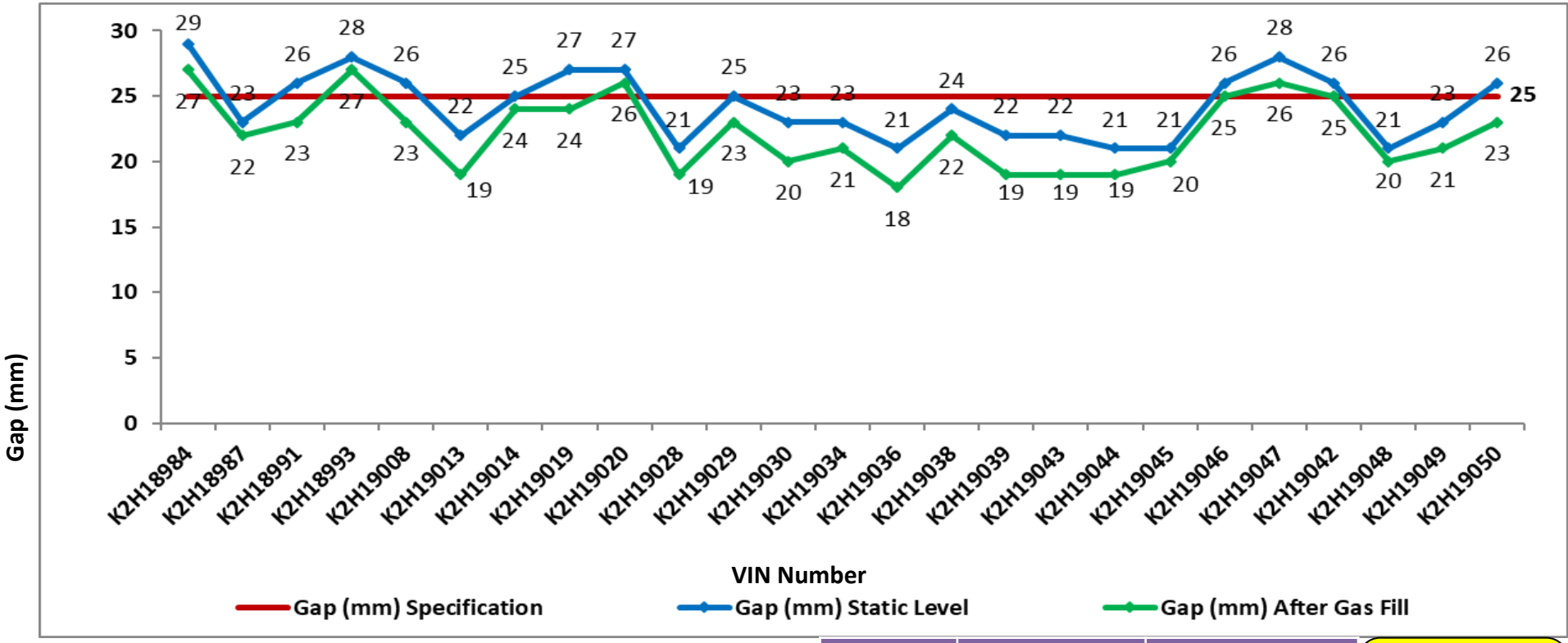
7.1 Collect data

Data of after action – Gap measured between hose and air cleaner

CAD clearances in existing layout is 25mm

7.1 Collect data

Data of after action – Gap measured between hose and air cleaner



	Static Level	After Gas Fill
Average	24.2	22.2
Min	21	18
Max	29	27

After Gas Fill Gap is 18 to 27 mm

Inference: **After change in bending angle hose gap increased from minimum 4 to 18mm.**

7.2 Capture all the effects

Effects on the selected problem:

- Rejection cost of 0.99 lac is saved annually
- 3, 6 & 12 MIS will improve

Any other positive effects observed:

- Customer (external) complaint resolved.
- Customer delight about Scorpio will increase.

Negative effects observed:

- NIL

7.3 Compare Before and After

Hypothesis testing for confirmation of results after PCA actions taken :-

Practical statement –

Defect rate of discharge hose cut(Y) was reduced after taking corrective measures through fixture correction and design actions.

Statistical Problem –

Ho: Defect rate after corrective measure >= Defect rate before corrective measure.

Ha: Defect rate after corrective measure < Defect rate before corrective measure .

Analysis: 2P Test -

7.3 Compare Before and After

Analysis: 2P Test

	Total vehicle produce	No of defects generated
Before action (Group 1)	4584	135
After action (Group 2)	9723	0

Statistical Conclusion: Reject the null hypothesis as P value observed is < 0.05
Practical Conclusion: Defect rate of door glass seal fold (Y) was eliminated after taking corrective actions at Supplier

7.3 Compare Before and After

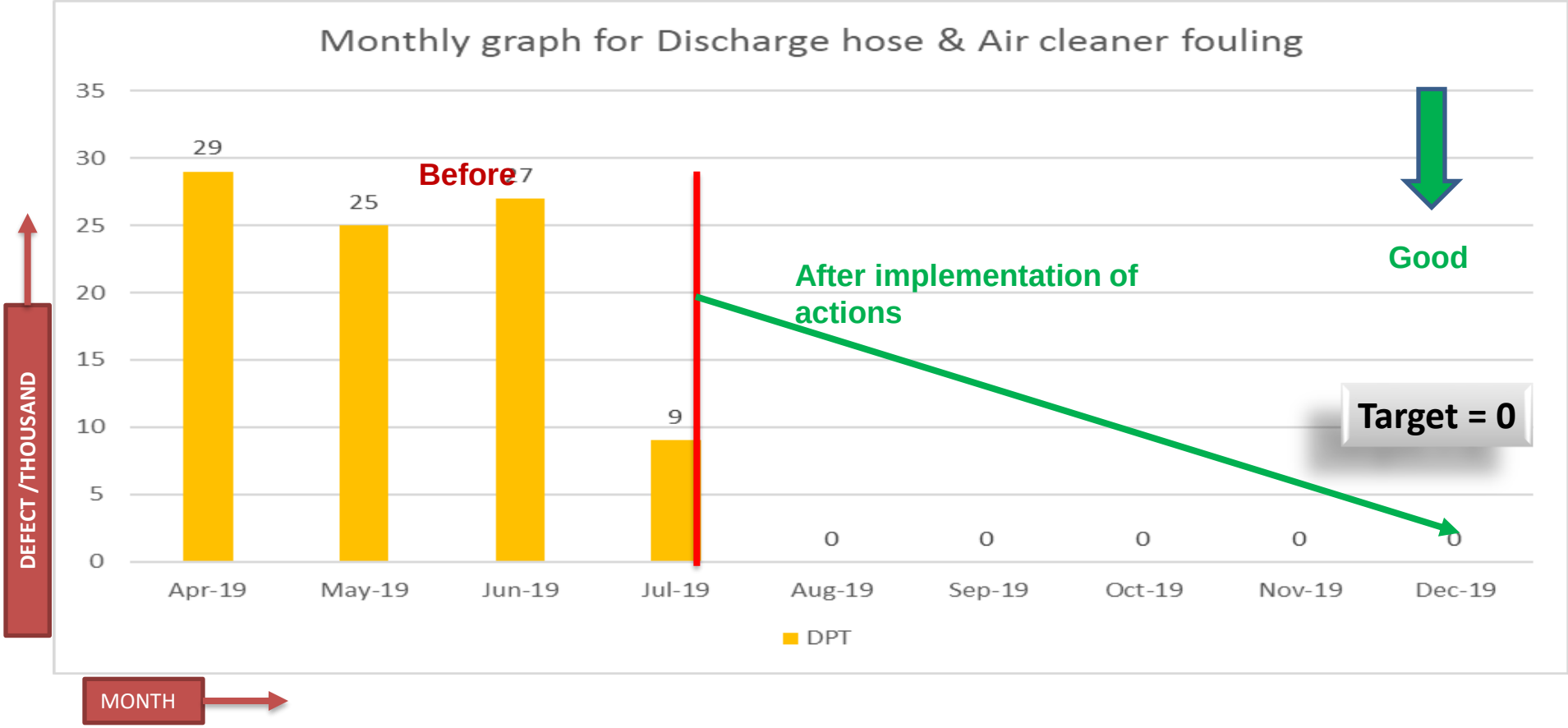
Data of after action – Sample size selection

	Total vehicle produce	No of defects generated
Before action (Group 1)	4584	135
After action (Group 2)	9723	0

Inference: **We have monitored 9723 vehicle and found OK**

7.3 Compare Before and After

Data of after action – Month wise defect trend



Inference: **Discharge hose fouling to air cleaner defect has been eliminated after implementation of corrective actions**

7.4 Measure Benefits

Tangible Benefits:

No	Details	Cost in Rs
1	**Cost of poor quality per vehicle** a) Rework cost: Rs.330/-	Rs. 330/-
2	**Material saving or any other cost benefits per vehicle (Rejection)**	Rs. 21344.82/-
3	**Productivity Benefits /any other benefits per vehicle**	
4	**No of vehicles produced per year (across sector) x Total Cost benefits per vehicle**	**Rs. 99239.28/-**

Intangible Benefits:

- Customer moral improved
- Dealer delight

8. Standardize

1. Update Documents
2. Train people to follow new standards
3. Audit standards

8.1 Update Document

AC discharge pipe fouling to air cleanerY):

S N	Document	Target date	Responsibility	Status
1	DRF to stage associate for Awareness.	24.10.2019	Rahul Gadekar	Closed
2	Supplier corrective action plan for pipe bending	21.06.2019	Nirmal Smabre	Closed
3	QA matrix for defect detection is updated.	24.10.2019	Rahul Gadekar	Closed
4	PFMEA updated for failure mode	24.10.2019	Rahul Gadekar	Closed
5	Buy off check sheet updated.	01.04.2019	Vikrant Ghule	Closed
6	Marking Standard updated for defect checking marking on buy off.	01.04.2019	Rahul Gadekar	Closed
7	EN release for pipe design change	09.10.2019	Rayappan Anandhan	Closed
8	Drawing updation for pipe profile change	09.10.2019	Rayappan Anandhan	Closed

8.2 Train people to follow new standards

8.2 Training & Awareness to operators & Inspectors (Supplier) -

Sr.No	Action Decided	Resp.	Actions taken	Status
1	OPL given to CAI & supplier associate to detect defect.	Sumit Mahale	Training & Awareness given to supplier stage operators OPL given to CAI assembly associate	**Closed**
2	V1 story sharing	Rahul Gadekar	V1 story shared through mail to TCF officers	**Closed**

8.3 Audit Standards

Sr.No	Action Decided	Resp.	Actions taken	Status
1	IQ/Process Audit check sheet to be updated for check point	Bhamare Sachin	IQ/Process Audit check sheet updated for check point	**Closed**

9. Summarize and Future Actions

1. Document Learning
2. Horizontal deployment
3. Future action

9. Summarize and Future Actions

9.1 Lessons Learnt :

1. Involvement of PVT helped in easy concern analysis
2. Check point addition at inhouse stopped defect from going into the field
3. CFT formation from various department helped in exact root cause which was a combination
4. DOE tools are very easy tool to identify the root cause & find out solution in a effective manner.
5. Take inputs from similar assembly process ongoing at some other location

9.2 Horizontal Deployment :

1. Actions taken were horizontally deployed at W150, U325, Z101 & W501 project.

9.3 Future Actions :

1. Touching/critical gap study input given to design teams for future projects

Thanks !!!!!

- **Team members**
- **Mentors** : Mr.Rahul Narkar,Mr.Shrikant Mulay.
- **MGB coordinator of commodity**: Mr.Sachin Bhamare
- **MGB Coordinators**: Mr.Ajay Namjoshi.
- **HR for providing learning opportunity**.
- **MQS team**
- **MQA team**